



Wrocław University
of Science and Technology

Design of a watercooled Solar Collector System

Solar Energy Conversion Systems — Course project

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1 INTRODUCTION

In the pursuit of sustainable energy solutions, this academic project is aimed at the design and simulation of a solar collector system. The project's core objective lies in the creation of a solar collector that meets specific requirements, such as the time frame and heating power needed, and the integral design considerations that derive from crafting a solution which optimizes efficiency and cost.

In harnessing solar radiation from the sun, the thermal solar collector arises as one of the best solutions (among other types such as photoelectric collectors), especially in large scale projects like this one. Despite the diverse array of solar collector types, they share a fundamental concept — a specialized material designed to capture and focus solar energy, ultimately transforming it into useful energy.

This section further lays the groundwork by providing an introductory exposition to solar collectors, offering a concise but comprehensive overview, crucial for understanding the decisions taken in the design and the calculations further ahead. In the subsequent section the specifications for this particular project will be exposed.

1.1 THEORIC BASIS

Solar collectors can be used for generating electricity through turbines or equivalents, or for the direct use of the collected heat – i.e. for domestic hot water and other purposes. This will be irrelevant for this project, as it will only concern on collecting a determinate amount of power as heated water, and will leave the rest of the system to the reader's consideration.

The components comprising a solar collector at its core are, ordered by layer, a cover, an absorber plate, a heat removal system, insulation and a case.

- **Cover:** Radiation-transmitting, generally made of glass or similar materials, prevents dust, dirt and other particles from reaching the inside components, mainly the absorber plate, while letting radiation pass through. It is important for this layer to be as transparent as possible in order to maximize the amount of radiation that reaches the collector, and it needs to be cleaned periodically once installed in order to avoid a decrease in efficiency.
- **Absorber plate:** At the core of the collector, typically fabricated from thermally conductive materials, serves as the primary interface for capturing incident solar radiation, heating up in the process.



- **Heat removal system:** Usually a fluid system which can consist on fins, pipes or other depending on the fluid at use. In most cases, water is used, usually travelling through pipes that heat up in contact with the absorber plate, removing consequently the heat from the plate and transporting it to wherever desired (i.e. the turbines above mentioned...).
- **Insulation:** A thick insulation layer isolates the hot parts of the collector – mainly the absorber plate and the heat removal system – from the outside environment, preventing heat loss and maximizing the useful heat obtained from the system.
- **Case:** Closes and encompasses all the systems above mentioned into a single, solid component granting it ease of installation, strength and protection from outside conditions.

All of it's elements allow the collector panel to collect radiation and transform it into useful heat. Further ahead, much more specific details about the design and aspect of individual parts, as well as the design and manufacturing methods for each one will be exposed, but the main elements of the system will remain the same.

1.2 SPECIFICATIONS

A set of specifications have been given for the realization of this particular project (topic 30). These comprise:

- **Liquid** type collector.
- Heating capacity: **24 kW**
- Final water temperature: $49^{\circ}C$
- Time frame of collection: **May 1st to September 30th**

Also, other specifications that remain the same for all projects include: the panels must be of size $1m \times 2m$. To reach the total power capacity multiple panels will need to be installed depending on the capacity of each panel until the total capacity demand is reached.



2 DESIGN

Having previously laid out the basic principles of functioning of a solar collector, this section will specify a particular design for the installation based on the specifications needed on this project. First, the design of single panels and its intended manufacturing process will be presented and specified, and later the general rules for the integration of different panels in a installation and how the connections should be made will be explained.

2.1 PANEL PARTS AND MATERIALS

The internal design of every element of the panel is specified in detail in this section, aswell as the materials and the manufacturing process, so every part can be manufactured from standard materials or parts. A typical thermal solar panel comprises several essential components: an absorber plate responsible for heat absorption, a fluid-based heat removal system (utilizing air or water), insulation to prevent heat dissipation from the panel, and a transparent cover that serves to shield the absorber plate from environmental influences. These elements are encapsulated within a casing, providing structural integrity to the entire system.

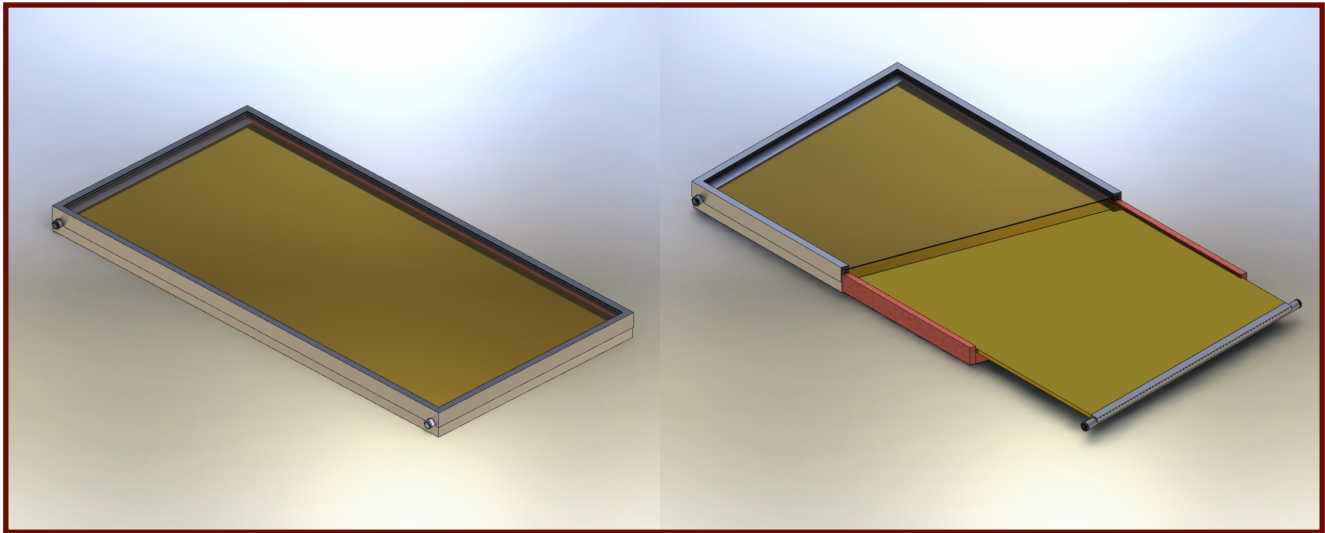


Figure 1. Renders of the solar panel closed and sectioned through the case and the insulation, respectively.

The adopted approach in this project diverges from the conventional method, incorporating a system where the **heat removal system and absorber plate have been unified** into a single, cohesive element. This modification significantly enhances efficiency, streamlining the design by eliminating problems associated with welds and multiple components. Consequently, the system



has an easier installation, needing less labor and eliminating the welding process. This, however, significantly complicates production, as the absorber is now a custom part that needs to be casted directly.

That is why panel design is explicitly oriented towards **mass production**, with the expectation that the per-unit cost will be substantially lower due to reduced labor requirements in the manufacturing process. However, it is essential to note that a considerable upfront investment is necessary for the production of specialized components, contrasting with traditional panels that commonly use readily available standardized parts.

2.1.1 COVER

The cover of the panel must be made from a material as transparent as possible, as this heavily impacts the efficiency of the collector. The considered options have been regular, high quality glass and PMMA. Polymethyl methacrylate (PMMA), often known by the trade name Plexiglas or acrylic, and glass are both materials commonly used for solar collector covers. Each material has its own set of advantages and disadvantages, and the choice between PMMA and glass depends on specific application requirements.

While PMMA has a higher transparency than glass, it degrades with time and often ends up being much less transparent, and has to be changed. It is desirable for panels to be durable and require little to no maintenance, thus **Silica Glass has been chosen for the cover**, justifying its additional cost. The cover will be 4mm thick, providing glass with enough resistance to support its own weight and resist breakage from unpredicted factors such as heavy ice rain, objects, etc...

<i>Cover: Glass</i>		
Thickness: δ_{cover}	4	mm
Heat Conductivity λ_{cover}	0,8	W/mK
Emissivity: ϵ_c	0,88	-
Refraction index: n_2	1,526	-

Table 1. Useful properties of the cover material: Glass

The glass will be mounted on the case and sealed with a rubber gasket. It will prevent air from flowing inside or outside of the panel, thus reducing heat loss.

2.1.2 ABSORBER BLOCK

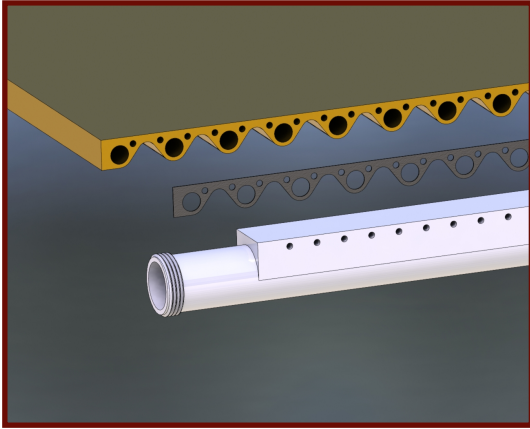


Figure 2. Exploded render of the Absorber block, the seal Gasket and the pipe that connects the inner tubes.

As said before, the absorber block and the heat removal piping system has been integrated into a single component. This component includes the refrigeration tubes inside the absorber plate, therefore making it as efficient as possible because heat transfers directly onto the flowing water. The block has 32 cooling tubes, the diameter of which will be calculated further ahead (Section 3.5).

When selecting a material for the absorber block, the primary consideration revolves around achieving a high heat conduction coefficient. This characteristic is crucial to ensure the efficiency of the absorber plate in effectively transferring heat from the absorbing surface at a higher temperature to the refrigerating tubes operating at a lower temperature.

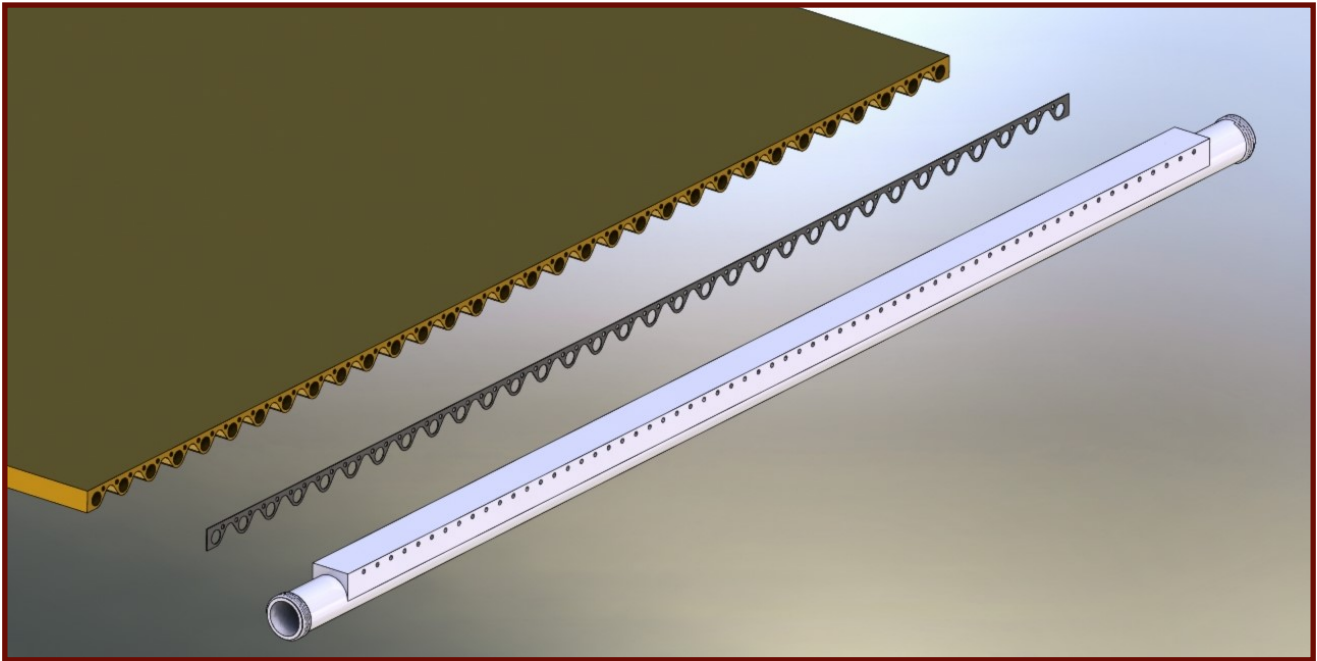


Figure 3. Exploded Render of the absorber system components. Note the absorber block's color does not reflect its material, but has been chosen for clearer representation.

Among the viable materials, metals conform most suitable options, with silver, copper, and aluminum exhibiting notable thermal conductivity. Silver, however, has been disregarded due to



its comparable thermal conductivity to copper but at a considerably higher cost. Consequently, the focus narrows down to copper and aluminum.

Copper boasts a thermal conductivity of approximately $400W/mK$, while aluminum registers around $250W/mK$. Considering the cost factors, with average prices of $8.3\$/kg$ for copper and $2.2\$/kg$ for aluminum, it becomes evident that, despite its superior thermal properties, copper may not justify its price. Consequently, the decision has been made to opt for **aluminum as the material for the absorber block**.

The plate is 4mm thick at its thinnest points in order to ensure the correct mechanical resistance of the block.

<i>Absorber Plate: Aluminum</i>		
Thickness: δ_{abs}	4	mm
Conductivity: λ_{abs}	250	W/mK

Table 2. Useful properties of the absorber plate material: Aluminum

2.1.3 PIPE ACCESSORIES

The absorber block is connected to two big pipe connectors on each end, which are sealed with two gaskets and pressed against the block with bolts. The pipe connectors consist of an extruded steel block with a big 25mm pipe which flows perpendicular to the cooling tubes and has drilled holes on its side that connect these cooling tubes to the big pipe. This big pipe has both its ends extended to emerge outside of the case, where a thread permits the connection with other pipes or a cap in order to connect the panel to the water circulation system in a variety of configurations.

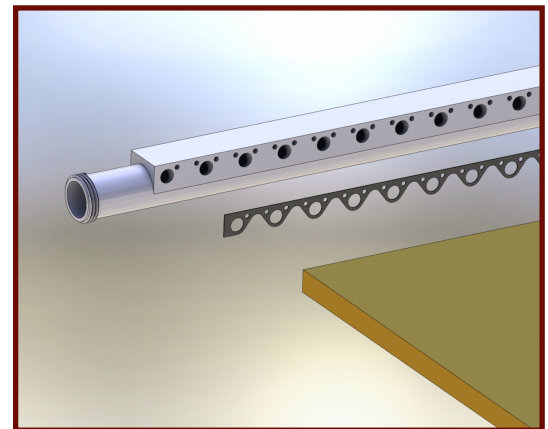


Figure 4. Exploded render of the Pipe connector, the seal Gasket and the absorber block, left to right.

The pipe is made of stainless steel that, while thermally conductive, conducts heat much worse than aluminium ($20W/mK$ vs. $250W/mK$), which is advantageous for its application, given that the pipe connects directly to the exterior and we want to minimize the heat losses as much as possible. The exterior piping, however, should always be insulated with foam or similar in order to minimize heat loss.



The gasket can be made either from gasket paper or rubber type sealers, it is about 0.5mm thick with the same section as the absorber plate.

2.1.4 COATING

An absorbing coating layer of black copper will be applied to the absorber block in order to reduce reflection and maximize the energy absorbed. **Black copper has been chosen** because of its superior absorbtance proprieties in comparison to other coatings.

<i>Coating: Black Copper</i>		
Emissivity: ϵ_p	0,18	-
Absorbtance: α_n	0,85	-

Table 3. Useful properties of the Coating material: Black Copper

2.1.5 INSULATION

Different types of insulators can be used in order to reduce heat losses due to convection in the bottom and walls, primarily: foams and wools. Foams are generally cheaper, but degrade over time and lose insulating proprieties, and can be susceptible to humidity damage, which is a possible working condition given that there will be water circulating inside the collector. Wools, on the other hand, do not absorb the humidity and generally don't degrade over time, which makes them better suited for this application. They are, however, more expensive in general.

For this project, **regular wool has been found to be the best material**. It has been chosen between mineral and glass wools, which all have similar proprieties regarding insulation capacity, working temperature and price.

The insulation is 30mm thick below the pipes and 48,75mm thick below the absorber, the most crucial part. On the walls, it is 25mm thick, as they have less surface area in contact with the absorber and are therefore, less important to cover.

<i>Insulation: Wool</i>		
Thickness: δ_{ins}	48,75	mm
Conductivity: λ_{ins}	0,04	W/mK

Table 4. Useful properties of the insulation material: Wool

2.1.6 CASE

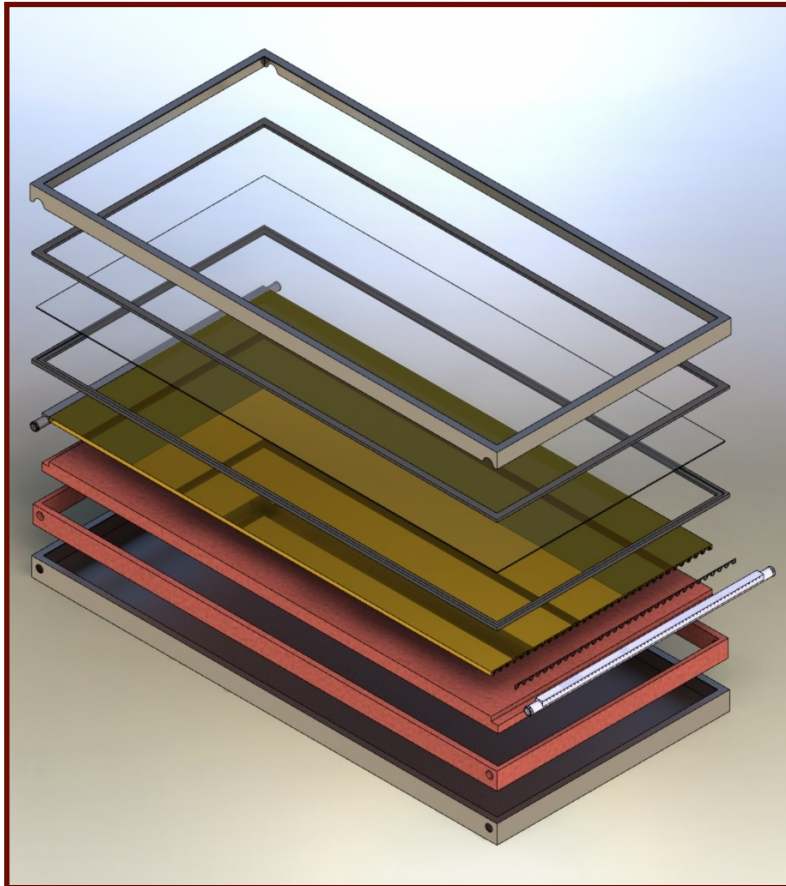


Figure 5. Exploded render of all the components in the panel, being the lower case the bottom component and the cover holder the top one.

The case is what encases all the components of the collector together, making them into a single component which absorbs the heat but remains insulated reducing losses. It has two components, the lower case and the cover holder. The lower case holds the insulation and the absorber and pipe system, while the cover is held within its seals between the lower case and the cover holder. The cover holder is then riveted to the lower case, enclosing all the components inside and converting the panel into a single component.

It is made from a 3mm plate of steel which is then bent to its final shape. Bending is a cheap manufacturing process that is ideal for mass production.

<i>Case: Stainless Steel</i>		
Thickness: δ_{case}	3	mm
Conductivity: λ_{case}	16	W/mK

Table 5. Useful properties of the case material: Stainless Steel



3 CALCULATIONS

3.1 AVERAGE RADIATION

To calculate the correct amount of energy the collector needs to absorb, it is assumed that the average amount of radiation the installation will receive is half of the peak radiation on an average day of the period. In order to calculate this, simulator software [SolarSym](#) has been used.

The first thing done has been inputting the time period and the place of the installation (Wrocław, Poland) in order to simulate the solar radiation for each day. From there, the average radiation/day in the time period has been calculated, resulting in $1052,09 \frac{W}{m^2}$. The day with the closest total radiation has been chosen as the simulated average day, this being day 232 (August 20th) with a total radiation of $1053 \frac{W}{m^2}$.

Another simulation has been later run to calculate the peak radiation received during the average day (day 232). The simulation has been set to give results for intervals of 5 minutes, and the peak measurement has been at time 12 : 30 PM resulting in $1067 \frac{W}{m^2}$. Thus, the average amount of radiation received is $1067 / 2 = 533,5 \frac{W}{m^2}$. This value will be considered for both the direct radiation (G_b) and diffuse radiation (G_d) This simplification was assumed in order to simplify the calculations further ahead:

$$G_b = 533,5 \frac{W}{m^2}$$

$$G_d = 533,5 \frac{W}{m^2}$$

3.2 DENSITY OF SOLAR RADIATION AND ANGLES

Next, the total entering heat radiation to the panel has been calculated. Firstly, all the relevant angles for the calculations have been calculated. These are vital for the correct installation of the system and the accurate calculation of the heat absorption and efficiency, and include the following values:

3.2.1 DECLINATION: δ

The declination angle refers to the angular measurement of the tilt of Earth's axis relative to the plane of its orbit around the Sun, influencing the variation in solar radiation received at a specific



location throughout the year. It depends on where the earth is in its orbit around the sun, so it can be calculated from the corresponding day of the year (in this case, day $n = 232$) as follows:

$$\delta = 23.45 \sin\left(360 \cdot \frac{284 + n}{365}\right) = 23.45 \sin\left(360 \cdot \frac{284 + 232}{365}\right) = 12.1^\circ$$

Where 23.45 is the tilt of earth axis in degrees.

3.2.2 LATITUDE: ϕ

Latitude is the angle a location forms north or south with Earth's equator. For the site of the instalation (Wrocław, Poland) the latitude is $\phi = 51.1^\circ$

3.2.3 INCIDENCE ANGLE: θ_β

The incidence of solar radiation angle refers to the vertical angle at which sunlight strikes the surface of the panel, measured between the incoming solar beam and a line perpendicular to the surface. From now on, it will be assumed that $\theta_\beta = 0^\circ$ for the rest of the calculations, meaning the panel is correctly oriented so that light hits it perpendicularly at the average time of the day.

3.2.4 HOUR ANGLE: ω

Hour angle is the measure of time since solar noon, expressed in angular distance along the celestial equator, providing the position of the sun relative to a specific observer's meridian. The hour angle at the peak hour (12:30h, in decimal 12.5h) can be calculated as follows because the sun moves laterally by 15° approximately every hour.

$$\omega = 15 \cdot (\tau - 12) = 15 \cdot (12.5 - 12) = 7.5^\circ$$

3.2.5 AZIMUTH ORIENTATION: γ

The ideal azimuth (N-S) orientation for the collector is always true south, as it is the average angle the sun's position has in a day, thus $\gamma = 0^\circ$.

3.2.6 ANGLE OF THE COLLECTOR β

The angle the collector has with the horizon. This can be calculated by solving the following formula for β , which will calculate the optimal vertical orientation for the collector.



$$\begin{aligned} \cos \theta_\beta &= \sin \delta \cdot \sin \phi \cdot \cos \beta - \sin \delta \cdot \cos \phi \cdot \sin \beta \cdot \cos \gamma + \cos \delta \cdot \cos \phi \cdot \cos \beta \cdot \cos \omega \\ &+ \cos \delta \cdot \sin \phi \cdot \sin \beta \cdot \cos \gamma \cdot \cos \omega + \cos \delta \cdot \sin \beta \cdot \sin \gamma \cdot \sin \omega \end{aligned} \quad (1)$$

$$\begin{aligned} \text{For: } \delta &= 12.1^\circ, \quad \phi = 51.1^\circ, \quad \theta_\beta = 0^\circ, \quad \omega = 7.5^\circ, \quad \gamma = 0^\circ \\ \beta &= 38.89^\circ \end{aligned}$$

3.2.7 SOLAR RADIATION DENSITY: G_β

Radiation density is the amount of solar energy per unit area received from the sun at a particular location and time, while oriented in the angles before stated. It can be calculated by using the following formula:

$$G_\beta = G_b \cdot R_b + G_d \cdot R_d + (G_b + G_d) \cdot R_0 \cdot \rho_0 \quad (2)$$

Where R_b and R_d are correction coefficients for the direct and diffuse radiations recieved (calculated in [3.1](#)) and R_0 is the correction coefficient for the reflected radiation and ρ_0 is the coefficient of reflection. Suposing the floor is made of grass we can take $\rho_0 = 0.25$. These coefficients can be calculated as:

$$\begin{aligned} R_b &= \frac{\cos(\phi - \beta) \cdot \cos \delta \cdot \cos \omega + \sin(\phi - \beta) \cdot \sin \delta}{\cos \phi \cdot \cos \delta \cdot \cos \omega + \sin \phi \cdot \sin \delta} \\ &= \frac{\cos(51.1^\circ - 38.89^\circ) \cdot \cos 12.1^\circ \cdot \cos 7.5^\circ + \sin(51.1^\circ - 38.89^\circ) \cdot \sin 12.1^\circ}{\cos 51.1^\circ \cdot \cos 12.1^\circ \cdot \cos 7.5^\circ + \sin 51.1^\circ \cdot \sin 12.1^\circ} \\ &= 1.2849 \end{aligned} \quad (3)$$

$$R_d = \frac{1 + \cos \beta}{2} = \frac{1 + \cos 38.89^\circ}{2} = 0.8892 \quad (4)$$

$$R_d = \frac{1 - \cos \beta}{2} = \frac{1 - \cos 38.89^\circ}{2} = 0.1108 \quad (5)$$

Then, substituting the coefficients in [2](#), the solar radiation density is obtained:

$$G_\beta = 533.5 \cdot 1.2849 + 533.5 \cdot 0.8892 + (533.5 + 533.5) \cdot 0.1108 \cdot 0.25 = 1189.43 \frac{W}{m^2}$$



3.3 ABSORBED RADIATION

In this section, a derivation process will be undertaken to transition from the calculated direct and diffuse radiations to the quantification of absorbed radiation by the absorber plate. The calculations will take into consideration both the radiation absorbed by the cover glass, the refraction phenomena associated with the different radiations incident upon the panel and the correction coefficients for each type of radiation, calculated in the last section. The absorbed radiation can be calculated as follows:

$$S = G_b \cdot R_b \cdot (\tau\alpha)_b + G_d \cdot R_d \cdot (\tau\alpha)_d + (G_b + G_d) \cdot \rho_0 \cdot R_0 \cdot (\tau\alpha)_0 \quad (6)$$

Where $(\tau\alpha)_x$ are the transmittance-absorbance products for direct, diffuse and reflected radiation respectively. In order to calculate $(\tau\alpha)_x$ for each radiation a few intermediate calculations must be done first.

3.3.1 REFRACTION IN THE COVER

When incident radiations hit the cover, they undergo simultaneous reflection and refraction processes. To accurately calculate the absorbed radiation, it is necessary to consider this phenomenon, given its different impact on direct, diffuse, and reflected radiation due to distinct incidence angles. The governing principle for refraction is encapsulated by Snell's Law, which states:

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2) \quad (7)$$

Where n_1 and n_2 represent the refractive indices of the incident and transmitting media respectively, these being air ($n_1 = 1$) and glass ($n_2 = 1.526$), θ_1 and θ_2 denote the angles of incidence and refraction. Under the assumption of proper panel orientation, the incidence angle θ_1 of direct radiation will be 0° . In the case of diffuse and reflected radiation, the following approximations for the angles can be applied:

$$\theta_d = 59.86 - 0.1388 \cdot \beta + 0.001497 \cdot \beta^2 \quad (8)$$

$$\theta_{1,d} = 59.86 - 0.1388 \cdot 38.89 + 0.001497 \cdot 38.89^2 = 56.72^\circ$$

$$\theta_0 = 90 - 0.5788 \cdot \beta + 0.002693 \cdot \beta^2 \quad (9)$$

$$\theta_{1,0} = 90 - 0.5788 \cdot 38.89 + 0.002693 \cdot 38.89^2 = 71.56^\circ$$



With the known incidence angles, the corresponding refraction angles can now be calculated:

$$\theta_2 = \arcsin\left(\frac{n_1}{n_2} \sin(\theta_1)\right) = \arcsin\left(\frac{1}{1.526} \sin(\theta_1)\right) \quad (10)$$

$$\theta_{2,b} = 0^\circ \quad \theta_{2,d} = 33,22^\circ \quad \theta_{2,0} = 38,43^\circ$$

3.3.2 COMPONENTS OF RADIATION

Utilizing the obtained refraction angles, the subsequent computation is determining the parallel and perpendicular components of radiation. These artificial values are necessary for the upcoming calculations, and are calculated as follows:

$$r_{\perp} = \frac{\sin^2(\theta_2 - \theta_1)}{\sin^2(\theta_2 + \theta_1)} \quad r_{\parallel} = \frac{\tan^2(\theta_2 - \theta_1)}{\tan^2(\theta_2 + \theta_1)} \quad (11)$$

Direct radiation:	$\theta_1 = 0^\circ$	$\theta_2 = 0^\circ$	$\Rightarrow$	$r_{\perp} = 0,043$	$r_{\parallel} = 0,043$
Diffuse radiation:	$\theta_1 = 56,726^\circ$	$\theta_2 = 33,221^\circ$	$\Rightarrow$	$r_{\perp} = 0,159$	$r_{\parallel} = 0,000$
Reflected radiation:	$\theta_1 = 71,563^\circ$	$\theta_2 = 38,438^\circ$	$\Rightarrow$	$r_{\perp} = 0,338$	$r_{\parallel} = 0,056$

Where, θ_1 and θ_2 are the incidence and refraction angles, respectively, of each particular radiation (the calculations have been repeated three times, one for each).

3.3.3 TRANSMISSIVITY

After calculating the effects of refraction in the incidence angles, the transmissivity of the glass cover must be taken into account. Transmissivity refers to the ability of the glass to allow sunlight or solar radiation to pass through and reach the absorber beneath. It is a key optical property that affects the efficiency of a solar panel in converting sunlight into heat. High transmissivity is desirable because it allows more sunlight to be absorbed by the absorber, increasing the overall efficiency of the solar panel.

In this section, the total transmissivity of the cover will be calculated. In order to do so, the glass transmittance or absorption losses in the glass must be calculated. The glass transmittance is dimensionless and can be found using the formula:

$$\tau_{\alpha} = \exp\left(-\frac{K \cdot \delta}{\cos(\theta_2)}\right) \quad (12)$$



$$\tau_{\alpha,b} = 0,919$$

$$\tau_{\alpha,d} = 0,904$$

$$\tau_{\alpha,0} = 0,898$$

Where K is the extinction coefficient, δ the thickness of the cover ($\delta = 6mm$), and θ_2 the refraction angle. The extinction coefficient is part of the complex refractive index used to characterize the optical properties of materials like glass. In this case it has been assumed to be $K = 21$.

Parting from the transmittance, the total cover transmissivity τ can be calculated. It is another dimensionless value between 0 and 1 that specifically quantifies the fraction of incident light that successfully passes through the material. It will also be calculated for each one of the three radiations:

$$\tau = \frac{\tau_{\alpha}}{2} \left[\frac{1 - r_{\perp}}{1 + r_{\perp}} \cdot \frac{1 - r_{\perp}^2}{1 - (r_{\perp} \cdot \tau_{\alpha})^2} + \frac{1 - r_{\parallel}}{1 + r_{\parallel}} \cdot \frac{1 - r_{\parallel}^2}{1 - (r_{\parallel} \cdot \tau_{\alpha})^2} \right] \quad (13)$$

$$\text{Direct radiation: } r_{\perp} = 0,043^{\circ} \quad r_{\parallel} = 0,043^{\circ} \quad \tau_{\alpha,b} = 0,919 \quad \Rightarrow \quad \tau_b = 0,843$$

$$\text{Diffuse radiation: } r_{\perp} = 0,159^{\circ} \quad r_{\parallel} = 0,000^{\circ} \quad \tau_{\alpha,d} = 0,904 \quad \Rightarrow \quad \tau_d = 0,779$$

$$\text{Reflected radiation: } r_{\perp} = 0,338^{\circ} \quad r_{\parallel} = 0,056^{\circ} \quad \tau_{\alpha,0} = 0,898 \quad \Rightarrow \quad \tau_0 = 0,618$$

3.3.4 TRANSMITTANCE-ABSORBTANCE PRODUCT

Manufacturers often use anti-reflective coatings on the absorber surface to minimize reflection and maximize transmissivity. These coatings help reduce the amount of sunlight that bounces off the absorber, ensuring that more light is absorbed. In this particular project, as stated in Section [2.1](#), the anti-reflective coating at use is black copper, with an absorptance of $\alpha_n = 0.85$.

Another part of the absorptance that also needs to be considered is angle dependent absorptance, which depends on the incidence angle of each radiation, and can be approximated as shown in Equation [14](#). This value has to be calculated for all three radiations.

$$\left(\frac{\alpha}{\alpha_n} \right) = 1 + 2.0345 \cdot 10^{-3} \cdot \theta_1 - 1.99 \cdot 10^{-4} \cdot \theta_1^2 + 5.324 \cdot 10^{-6} \cdot \theta_1^3 - 4.799 \cdot 10^{-8} \cdot \theta_1^4 \quad (14)$$

$$\left(\frac{\alpha}{\alpha_n} \right)_b = 1,000 \quad \left(\frac{\alpha}{\alpha_n} \right)_d = 0,950 \quad \left(\frac{\alpha}{\alpha_n} \right)_0 = 0,819$$

Now, all the data needed to calculate the transmittance-absorptance product is available. The transmittance-absorptance product gives the fraction of incident light energy that is neither transmitted nor reflected. Instead, it is absorbed by the material, in this case, the absorber plate. The product can be calculated as follows:



$$(\tau\alpha) = 1.01 \cdot \tau \cdot \left(\frac{\alpha}{\alpha_n} \right) \cdot \alpha_n \quad (15)$$

$$\text{Direct radiation: } \tau_b = 0,843 \quad (\alpha/\alpha_n)_b = 1,000 \quad \Rightarrow \quad (\tau\alpha)_b = 0,724$$

$$\text{Diffuse radiation: } \tau_d = 0,779 \quad (\alpha/\alpha_n)_d = 0,950 \quad \Rightarrow \quad (\tau\alpha)_d = 0,635$$

$$\text{Reflected radiation: } \tau_0 = 0,618 \quad (\alpha/\alpha_n)_0 = 0,819 \quad \Rightarrow \quad (\tau\alpha)_0 = 0,434$$

3.3.5 ABSORBED RADIATION

Finally, the absorbed radiation can be calculated using Equation [6](#). Now that all of the values in the formula have been calculated, the total absorbed radiation by the absorber plate is:

$$\begin{aligned} S &= G_b \cdot R_b \cdot (\tau\alpha)_b + G_d \cdot R_d \cdot (\tau\alpha)_d + (G_b + G_d) \cdot \rho_0 \cdot R_0 \cdot (\tau\alpha)_0 = \\ &= 533.5 \cdot 1.2849 \cdot 0.724 + 533.5 \cdot 0.8892 \cdot 0.635 + (533.5 + 533.5) \cdot 0.1108 \cdot 0.25 \cdot 0.434 = \\ &= 810,100 \frac{W}{m^2} \end{aligned}$$

3.4 HEAT LOSS

In this section, the comprehensive assessment of total heat loss within the collector, attributable to convection, will be conducted. This encompasses the sum of heat dissipation through the collector's walls, the lower surface, and the glazing.

3.4.1 TEMPERATURES

To initiate the heat loss calculations, it is necessary to perform compute some intermediate values, starting with temperatures. These essential temperatures for subsequent analyses include T_{sky} , denoting the sky temperature; T_{psr} , representing the absorber plate temperature; and T_c , indicating the maximum temperature tolerance for conventional outdoor glass. They can be calculated as follows, respectively.

$$T_{sky} = 0.0552 \cdot T_a^{1.5} = 0.0552 \cdot 298^{1.5} = 284,179K \quad (16)$$

$$T_{psr} = \left(\frac{S}{\sigma \cdot \epsilon_p} \right)^{0.25} = \left(\frac{810,097}{5,67 \cdot 10^{-8} \cdot 0,18} \right)^{0.25} = 530,787K \quad (17)$$

$$T_c = \frac{(T_a + T_{psr})}{2} = \frac{(298 + 530,787)}{2} = 414,469K \quad (18)$$

Where T_a is the ambient temperature (taken as $25^\circ C$ in summer), S the absorbed radiation just calculated, σ The Stephan-Boltzmann constant ($5,67 \cdot 10^{-8} W/m^2 K^4$), and ϵ_p the absorber (coating) emissivity (Table 3, 0.18).

3.4.2 CONVECTION COEFFICIENTS

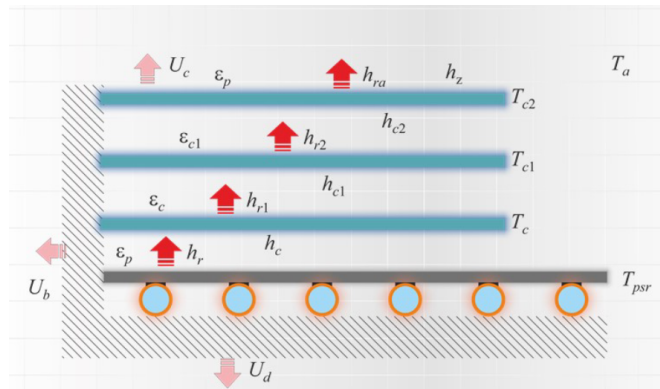


Figure 6. The convection coefficients represented graphically

With these temperatures, the following needed convection coefficients can be calculated: h_r ,



the convection coefficient; h_{ra} , the radiative heat loss coefficient; h_z , the convection coefficient for air flowing over the outside surface of the glass cover; h_1 and h_2 , the convention coefficients for the cover and absorber, respectively.

$$h_r = \sigma \cdot \frac{T_{psr}^2 + T_c^2}{T_{psr} + T_c} \cdot \frac{1}{\frac{1}{\epsilon_p} + \frac{1}{\epsilon_c} - 1} =$$

$$= 5,67 \cdot 10^{-8} \cdot \frac{530,787^2 + 414,469^2}{530,787 + 414,469} \cdot \frac{1}{\frac{1}{0,18} + \frac{1}{0,88} - 1} = 4,270 \text{ W/m}^2\text{K} \quad (19)$$

$$h_{ra} = \frac{T_{psr}^2 + T_c^2}{T_{psr} + T_c} \cdot \frac{T_{psr} + T_c}{\frac{1}{\epsilon_c} + \frac{1}{\epsilon_{sky}} - 1} \cdot \sigma =$$

$$= \frac{530,787^2 + 414,469^2}{530,787 + 414,469} \cdot \frac{530,787 + 414,469}{\frac{1}{0,88} + \frac{1}{1} - 1} \cdot 5,67 \cdot 10^{-8} = 21,390 \text{ W/m}^2\text{K} \quad (20)$$

$$h_z = 2.8 + 3 \cdot v_t = 2.8 + 3 \cdot 3 = 11,800 \text{ W/m}^2\text{K} \quad (21)$$

$$h_1 = 2.8 + 3 \cdot v_w = 2.8 + 3 \cdot 0.1 = 3,100 \text{ W/m}^2\text{K} \quad (22)$$

$$h_2 = 2.8 + 3 \cdot v_b = 2.8 + 3 \cdot 0.1 = 3,100 \text{ W/m}^2\text{K} \quad (23)$$

Where ϵ_c is the emissivity of the glass cover (Table 1), ϵ_{sky} the emissivity of the sky (1), v_t the velocity of the air over the cover surface, which has been assumed to be 3m/s , v_w and v_b the velocities of the air under and around the absorber, which have been assumed to be 0.1m/s (both).

3.4.3 PARTIAL HEAT LOSSES

The partial heat losses through the walls U_b , bottom U_d and cover U_c have been computed from all the data calculated previously:

$$U_b = \frac{A_{abs}}{A_w \cdot \left(\frac{1}{h_b} + \frac{\delta_{case}}{\lambda_{case}} + \frac{\delta_{ins}}{\lambda_{ins}} + \frac{1}{h_z} \right)} =$$

$$= \frac{1.805}{0.6 \cdot \left(\frac{1}{2.8} + \frac{3 \cdot 10^{-3}}{16} + \frac{48.75 \cdot 10^{-3}}{0.04} + \frac{1}{3.1} \right)} = 1.811 \text{ W/m}^2\text{K} \quad (24)$$



$$U_d = \frac{1}{\frac{1}{h_z} + \frac{\delta_{case}}{\lambda_{case}} + \frac{\delta_{ins}}{\lambda_{ins}}} = \quad (25)$$

$$= \frac{1}{\frac{1}{3.1} + \frac{3 \cdot 10^{-3}}{16} + \frac{48.75 \cdot 10^{-3}}{0.04}} = 0.767 \text{ W/m}^2\text{K}$$

$$U_c = \frac{1}{\frac{1}{h_c+h_r} + \frac{1}{h_z+h_{ra}} + \frac{\delta_{cover}}{\lambda_{cover}}} = \quad (26)$$

$$= \frac{1}{\frac{1}{2.8+4.27} + \frac{1}{3.1+21.39} + \frac{4 \cdot 10^{-3}}{0.8}} = 5.664 \text{ W/m}^2\text{K}$$

Where A_{abs} is the area of the absorber plate ($1.9m \cdot 0.95m = 1.805m^2$), A_w is the area of the walls ($2 \cdot 0.1 \cdot 2 + 2 \cdot 0.1 \cdot 1 = 0.6m^2$), h_b the convection coefficient for the walls ($h_b = 2.8 \text{ W/m}^2\text{K}$), δ_{case} and λ_{case} the thermal properties of the case material (Table 5), δ_{ins} and λ_{ins} the thermal properties of the insulation material (Table 4) and δ_{cover} and λ_{cover} the thermal properties of the cover material (Table 1).

3.4.4 TOTAL HEAT LOSS

The total heat losses in the collector amount to the sum of the partial losses formerly computed:

$$U_L = U_b + U_d + U_c = 1.811 + 0.767 + 5.664 = 8.242 \text{ W/m}^2\text{K} \quad (27)$$

3.5 USEFUL HEAT

With the total absorbed radiation and the heat losses, the useful heat collected by the water can be calculated. It measures how much heat the collector is able to extract from the absorber plate using the cooling system. The first thing to be calculated is the standard Fin efficiency F , and the collector efficiency factor F' , followed by the heat removal factor F_r . These steps will allow us to ultimately calculate the Useful Heat Q_u . In order to calculate these, though, the geometry of the absorber plate needs to be defined. In order to do so, the mass flow of the pipes must first be estimated, which can be done knowing how much the temperature increases in one panel.

As for the specifications of the project, the final temperature of the water is to be 49°C and the initial temperature has been taken to be 19°C as it's the stipulated temperature for running water in Poland. To maximize the efficiency of a heat exchanger it is ideal to have a temperature difference ΔT around 5°C in liquid exchangers, which will be achieved by connecting several panels (in this case, 5, making for a $\Delta T = 6^\circ\text{C}$) in series. Then, the mass flow can be calculated



from the obtained heat, which we will estimate as the absorbed radiation times the area of the absorber block in order to get a ceiling for how much the mass flow can be.

$$\dot{m} = \frac{S \cdot A}{C_{p,water} \cdot \Delta T} = \frac{810.097 \cdot 1.805}{4180 \cdot 6} = 0.0588 \text{ kg/s} \quad (28)$$

We assume $C_{p,water} = 4180 \text{ J/kgK}$. From this mass flow, calculations have been made to know the pressure loss in the collector for different diameters of pipes. This has been done using the internet tool [Online pressure drop calculator](#).

Calculation output Flow medium: Water 39 °C / liquid Mass flow: 0.076027779 kg/s Weight density: 998.206 kg/m³ Dynamic Viscosity: 1001.61 10-6 kg/ms Element of pipe: circular Dimensions of element: Diameter of pipe D: 22 mm Length of pipe L: 2 m Velocity of flow: 0.2 m/s Reynolds number: 4393 Velocity of flow 2: - Reynolds number 2: - Flow: turbulent Absolute roughness: 0.0015 mm Pipe friction number: 0.04 Resistance coefficient: 3.54 Resist. coeff. branching pipe: - Press. drop. branch pipe: - Pressure drop: 0.71 mbar 0 bar	Calculation output Flow medium: Water 39 °C / liquid Mass flow: 0.076027779 kg/s Weight density: 998.206 kg/m³ Dynamic Viscosity: 1001.61 10-6 kg/ms Element of pipe: circular Dimensions of element: Diameter of pipe D: 15.5647575 mm Length of pipe L: 2 m Velocity of flow: 0.4 m/s Reynolds number: 6213 Velocity of flow 2: - Reynolds number 2: - Flow: turbulent Absolute roughness: 0.0015 mm Pipe friction number: 0.04 Resistance coefficient: 4.54 Resist. coeff. branching pipe: - Press. drop. branch pipe: - Pressure drop: 3.63 mbar 0 bar	Calculation output Flow medium: Water 39 °C / liquid Mass flow: 0.076027779 kg/s Weight density: 998.206 kg/m³ Dynamic Viscosity: 1001.61 10-6 kg/ms Element of pipe: circular Dimensions of element: Diameter of pipe D: 9.838779148 mm Length of pipe L: 2 m Velocity of flow: 1 m/s Reynolds number: 9823 Velocity of flow 2: - Reynolds number 2: - Flow: turbulent Absolute roughness: 0.0015 mm Pipe friction number: 0.03 Resistance coefficient: 6.35 Resist. coeff. branching pipe: - Press. drop. branch pipe: - Pressure drop: 31.83 mbar 0.03 bar	Calculation output Flow medium: Water 39 °C / liquid Mass flow: 0.076027779 kg/s Weight density: 998.206 kg/m³ Dynamic Viscosity: 1001.61 10-6 kg/ms Element of pipe: circular Dimensions of element: Diameter of pipe D: 3.11129515 mm Length of pipe L: 2 m Velocity of flow: 10.02 m/s Reynolds number: 31063 Velocity of flow 2: - Reynolds number 2: - Flow: turbulent Absolute roughness: 0.0015 mm Pipe friction number: 0.02 Resistance coefficient: 15.79 Resist. coeff. branching pipe: - Press. drop. branch pipe: - Pressure drop: 7909.08 mbar 7.91 bar
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Figure 7. Pressure drop in one collector length for the diameters of pipe: 22mm, 15,5 mm, 9,8 mm, and 3mm respectively

As seen in Figure 7, The pressure drop gets exponentially significant the single digits of the diameter in mm are approached. This is why the final decided diameter of the pipes has been $D = 10 \text{ mm}$. The separation between pipes will be $W = 30 \text{ mm}$ in order to have 32 pipes in the absorber block, a bit more than what is normally implemented with $W = 2 \cdot D$ but given that there are no welds and the architecture of the absorber block has been designed specifically to maximize heat transfer, this will suppose no significant reduction of efficiency at the end.

The standard fin efficiency can now be calculated using formula 29

$$F = \frac{\tanh \left(\sqrt{\frac{U_L}{\lambda_{abs} \cdot \delta_{abs}}} \cdot (W - D) \cdot 0.5 \right)}{\sqrt{\frac{U_L}{\lambda_{abs} \cdot \delta_{abs}}} \cdot (W - D) \cdot 0.5} = \frac{\tanh \left(\sqrt{\frac{8.242}{250 \cdot 4 \cdot 10^{-3}}} \cdot (0.030 - 0.010) \cdot 0.5 \right)}{\sqrt{\frac{8.242}{250 \cdot 4 \cdot 10^{-3}}} \cdot (0.030 - 0.010) \cdot 0.5} = 1.000 \quad (29)$$

Where δ_{abs} and λ_{abs} are the thermal properties of the cover material (Table 2), W is the distance between tubes and D the diameter of the cooling tubes. The fin efficiency factor F' can be calculated as follows.



$$F' = \frac{1}{U_L \cdot W \cdot \left(\frac{1}{U_L \cdot (W-D) \cdot (1+F \cdot (W-D))} + \frac{\gamma}{\lambda_{abs} \cdot b} + \frac{1}{\pi \cdot d \cdot h_{fi}} \right)} = \quad (30)$$

$$= \frac{1}{8.242 \cdot 0.03 \cdot \left(\frac{1}{8.242 \cdot (0.03-0.01) \cdot (1+1 \cdot (0.03-0.01))} + 0 + \frac{1}{\pi \cdot 0.01 \cdot 90} \right)} = 0.919$$

Where h_{fi} is the convection coefficient of the pipes (estimated as $h_{fi} = 90W/mK$). Notice that the term $\frac{\gamma}{\lambda_{abs}}$ has been assumed to be 0 because there is no weld between the pipes, so $\gamma = 0mm$. This is one of the factors in which the unorthodox design of the absorber block improves efficiency dramatically.

The heat removal factor F_r is calculated subsequently:

$$F_r = \frac{\dot{m} \cdot C_{p,water}}{A_{abs} \cdot U_L} \cdot \left(1 - \exp \left(-\frac{U_L \cdot F' \cdot A_{abs}}{\dot{m} \cdot C_{p,water}} \right) \right) = \quad (31)$$

$$= \frac{0.058 \cdot 4180}{1.805 \cdot 8.242} \cdot \left(1 - \exp \left(-\frac{8.242 \cdot 0.919 \cdot 1.805}{0.058 \cdot 4180} \right) \right) = 0.894$$

With all the values calculated, the useful heat can be calculated finally, which considers the absorbed radiation, the heat losses of the panel and the heat that is able to be extrated by the water:

$$Q_u = A_{abs,plate} \cdot F_r \cdot (S - U_L \cdot \Delta T) = 1.805 \cdot 0.894 \cdot (810.097 - 8.242 \cdot 6) = \quad (32)$$

$$Q_u = 1227.567 \text{ W/panel}$$

3.6 COLLECTOR EFFICIENCY

The efficiency of the collector can be calculated using the total density of radiation, calculated in [3.2.7](#)

$$\eta = \frac{\text{Useful heat}}{\text{Total Radiation}} = \frac{Q_u}{G_\beta \cdot A_{abs,plate}} = \frac{1227.567}{1189.42} = 0.681 \Rightarrow 68.1\% \quad (33)$$

3.7 NUMBER OF COLLECTORS

The number of collectors can be calculated based on the total capacity needed by the project specifications and the useful heat obtained from each collector.

$$n = \frac{Q_{cap}}{Q_u} = \frac{24000}{1227.567} = 19.55 \Rightarrow 20 \text{ panels} \quad (34)$$

The panels will be installed in a 5 by 4 grid, with 4 series of 5 panels each in series, mounted in parallel, as shown in the image.

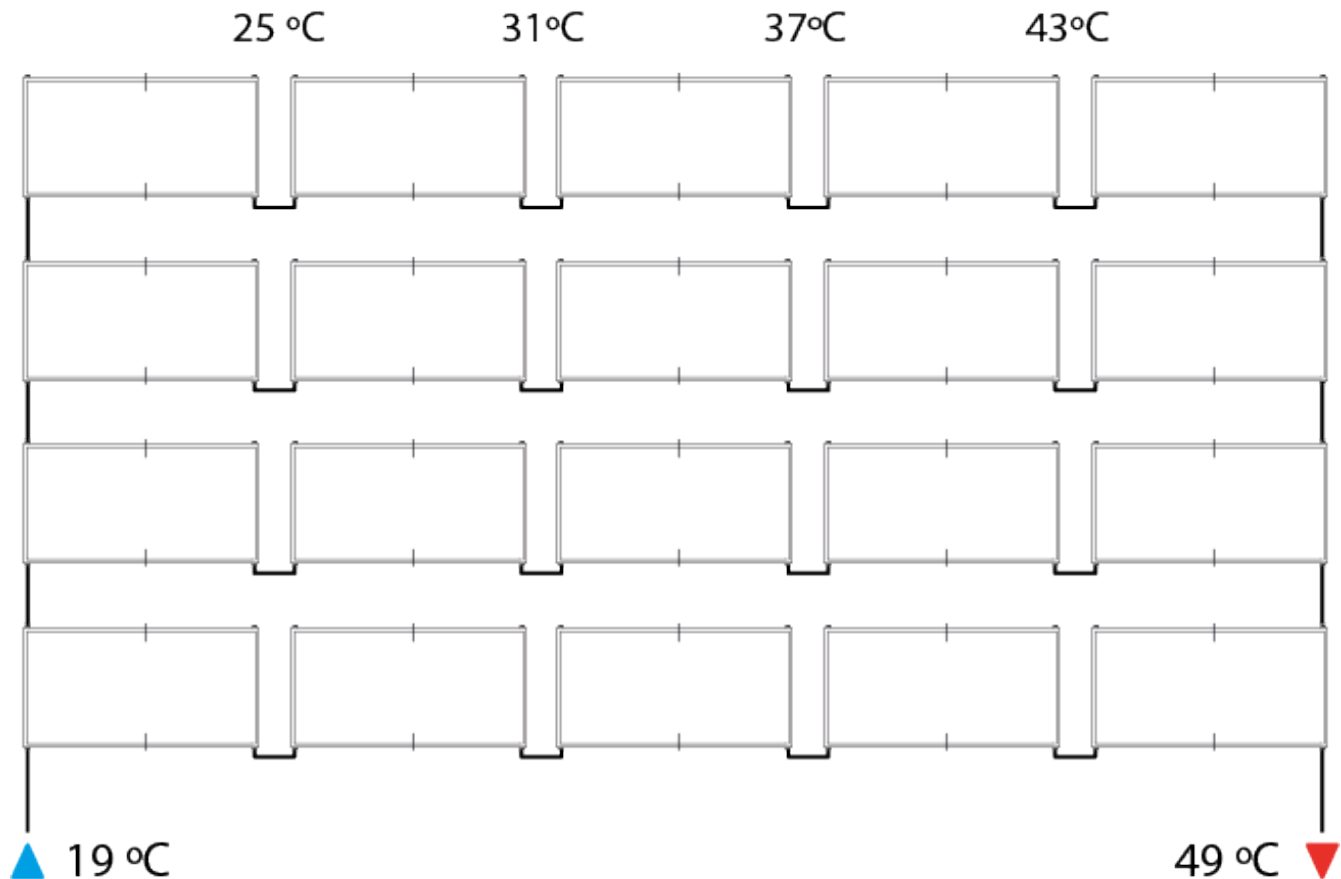


Figure 8. Diagram of the proposed panel connections for the installation. All the unused tubes of the panels are fitted with caps.



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4 ANNEX: DRAWINGS

D

C

B

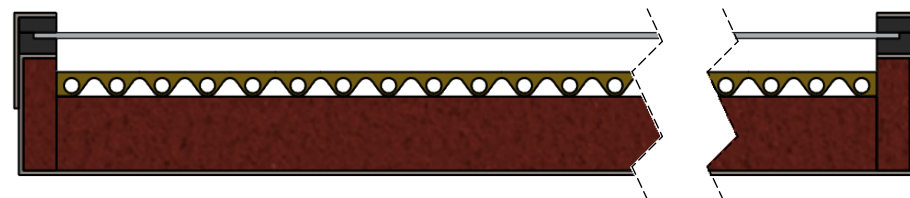
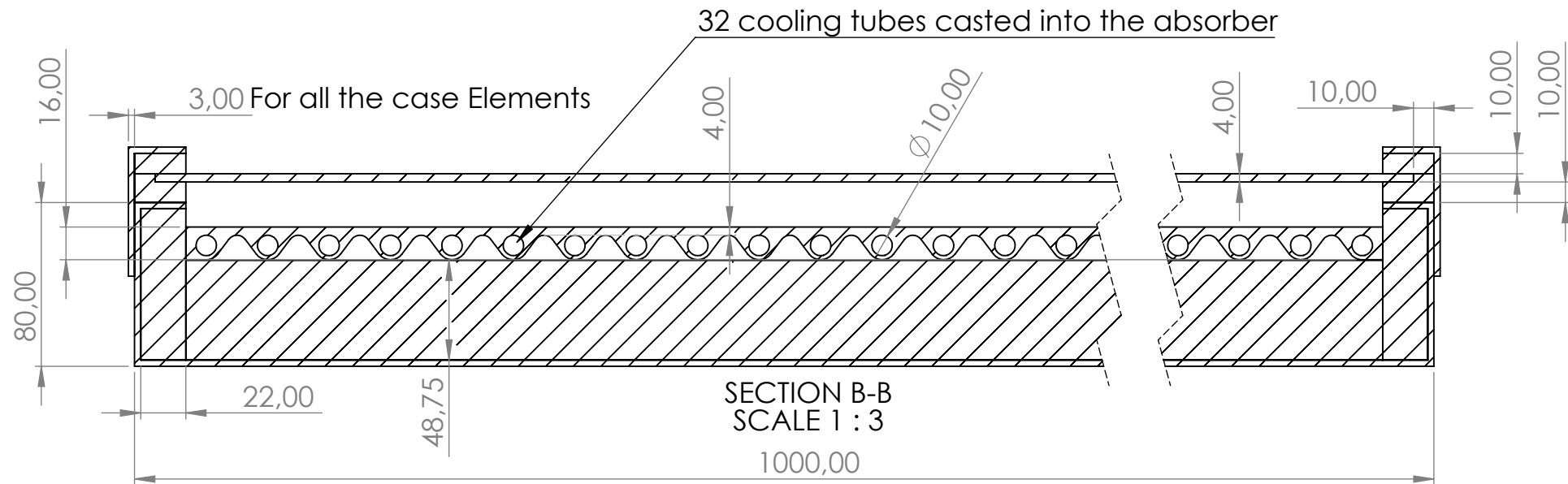
A

D

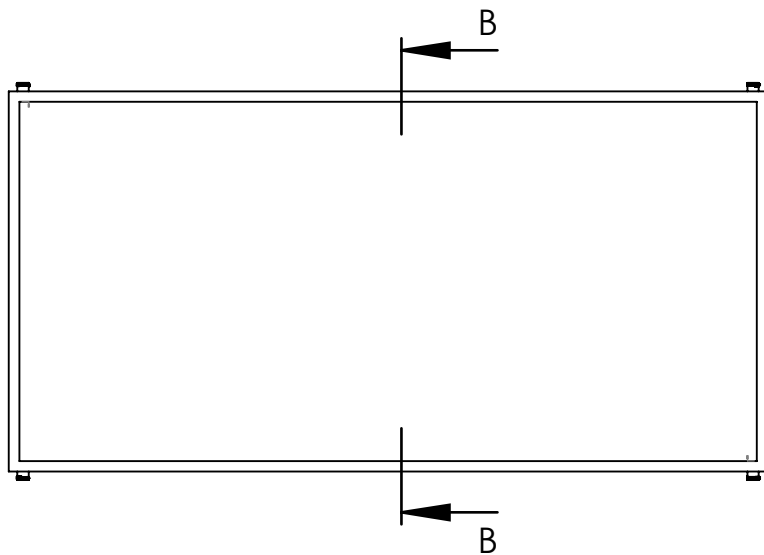
C

B

A



SECTION B
SCALE 1 : 5



NO CAMBIE LA ESCALA		REVISIÓN
TÍTULO:		
Panel Section		
N.º DE DIBUJO		A4
Solar Panel		
ESCALA 1:10		HOJA 1 DE 1

